Measuring Online Tracking with Ad Blockers and VPNs

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Abstract

This project examines VPNs and ad blockers as tools to minimize online tracking activities while using different browsing modes within the Chrome browser environment. The research tracked third-party requests, cookies, Javascript API calls, fingerprinting and cookie syncing across four conditions: vanilla, VPN only, adblock only, and VPN with adblock. The DuckDuckGo Tracker Radar collected data while the evaluation scope expanded to 540 website for complete assessment. We evaluated different VPN interface types by testing NordVPN with a Graphical User Interface (GUI) and ProtonVPN with a Command Line Interface (CLI). Our research demonstrates how privacy tools function during actual Chrome browsing sessions and identifies essential considerations for users who want to improve their online privacy.

1. Introduction

Modern web browsing includes online tracking as a fundamental yet concealed element. Websites receive data from multiple third-party entities through cookie collection and JavaScript API calls and fingerprinting and cookie syncing methods. Users protect their privacy by using Virtual Private Networks (VPNs) and ad blockers as countermeasures against these threats. The actual ability of these tools to minimize tracking remains untested in real-world browsing scenarios especially when used within a standard browser framework such as Chrome.

The goal of this project is to measure and evaluate the level of online tracking users experience under different privacy-enhanced browsing configurations. In particular, we want to know how third-party requests, cookies, fingerprinting activities, and cookie syncing differ across four browsing modes: vanilla (no protection), VPN only, adblock only, and VPN plus adblock.

The combination of VPN and ad blocker will reduce online tracking significantly compared to vanilla browsing. Additionally, we hypothesize that the VPN's deployment type (GUI vs CLI)[1] may affect the amount of tracking exposure due to the differences in background services and configurations. Our methods include automated crawling of 540 websites using the Chrome browser while systematically changing the privacy setup under test, and using the DuckDuckGo Tracker Radar to capture and analyze tracking data.

2. Background & Related Work

Online tracking research continues to be a dynamic field of investigation which focuses on website and third-party service data collection methods and user data exploitation. Research has shown that cookies and fingerprinting and other tracking technologies are used by a significant portion of websites.[2] OpenWPM and EFF studies have shown that basic browsing actions result in major data collection and profiling activities. [2][3]

The DuckDuckGo Tracker Radar serves as a significant resource for the privacy research community by tracking known trackers and their tracking behaviors. Tracker Radar has been employed in various independent research to measure tracker occurrences and their frequency throughout the internet.[4] Tracker Radar provides a more accurate view of tracking activity because it dynamically analyzes web behavior instead of using traditional adblock lists.[4]

Several studies have also investigated the effectiveness of VPNs and ad blockers in reducing tracking. While VPNs primarily serve to encrypt traffic and mask IP addresses, their ability to block trackers is limited unless combined with additional tools like tracker blockers or ad blockers.[5] Some studies have also noted that differences in VPN implementation such as background services, telemetry, and interface design can affect the level of privacy offered.[5]

In addition, previous studies show that many VPN providers are not immune to leaks of information such as DNS queries, traffic that sometimes eliminates anonymity completely even with active VPNs[6]. These technical limitations also emphasize the need to combine protections that rely on VPNs with browser-based protections in order to ensure that meaningful privacy is achieved.

The literature lacks research that directly compares VPNs through their interface type (GUI vs CLI) and assesses the impact of this difference on online tracking exposure. Our project fills this research gap by investigating whether the deployment method of VPNs (user-friendly apps versus manual command-line connections) affects user privacy.

Our research builds upon existing knowledge by adding this distinctive comparison to understand VPN architecture and privacy tool interactions with tracking resistance in Chrome browsing.

3. Methodology

In order to systematically assess the effectiveness of VPNs and ad blockers in reducing online tracking, we structured a typical experiment that offered four different browsing scenarios: Vanilla mode (i.e., no privacy-enhancing tools were enabled), VPN-only (i.e., VPN was enabled without ad-blocking software), Ad blocker only (i.e., ad-blocking extension was enabled without a VPN), and VPN with ad blocker (i.e., both VPN and ad-blocker were simultaneously operational). This experiment was intended to be a method for enabled us to assess and quantify how well two individual privacy mechanisms performed on tracking behavior of third-parties and how those same mechanisms worked together quantitatively and qualitatively. We gauged how well either a VPN or ad blocker worked against online tracking in a variety of browsing scenarios and environments with all configurations and combinations of each. [7]

3.1. Selection of Websites

Our research, and the prior research described above, advances by examining 540 websites drawn from the Tranco top websites chart, randomly sampled, which represented a sampling of sites across various environments. These websites represented a general sampling of popular web domains that enabled reasoning about a realistic range of online tracking practices.

3.2. Data Collection Procedure

We automated the data collection process with the DuckDuckGo Tracker Radar Collector. This is an open source data collecting tool used to detect and classify web trackers.[4] To further ensure that our approach was consistent and reproducible, we collected data after conducting a series of experiment tests in a controlled environment and by using the latest stable version of the Chrome browser for each experiment under the same technical circumstances; each test target was performed on a fresh installation of the browser. In reference to being controlled, browser caches, cookies and browsing histories were properly cleaned in orderly fashion between testing sessions so as to mitigate contamination across experimental conditions. Our data collection focused on specific tracking behaviors related to trackers previous web behaviours, more specifically third party HTTP(S) requests (requests from domains other than the one primarily visited), the creation and access of cookies, and JavaScript API calls introduced for fingerprinting, which includes the Canvas, WebGL, AudioContext, and Battery Status APIs. We also implemented public fingerprinting attempts as well as attempts for cookie synchronization reported across 3rd party domains. This reliable repeated procedure ensured that we were able to have thorough tracking data that will be reproducible for all experimental conditions.

3.3. VPN and Ad blocker Configuration

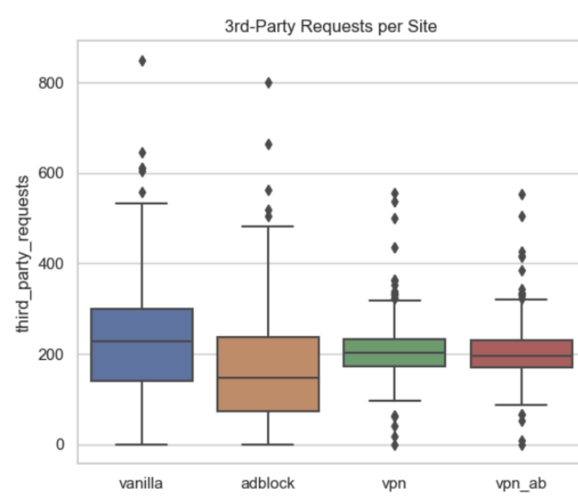
During our tests, we considered two common VPN products that each reflected a different type of user interface. We selected to analyze NordVPN via its GUI, representing a typical consumer experience. We selected ProtonVPN and tested it using a command-line interface (CLI) on a Linux-based platform, thus completely abstracting the GUI-components to be more apples-to-apples in the comparison.[1] We also considered the blocking tool uBlock Origin and, to some extent, used its default settings to simulate a common user use case with certain configurations. By using the ad-blocker in its default state in our configuration, we ensured our experimental configuration would broadly represent a user configuration as had both technologies been used by the user. This way we positioned the experiment to represent common use cases, and normalized the comparative distance between the capabilities of the VPNs, as well as the ad-blocker as it pertained to concealing users from probable agents of online tracking.[7]

4. Evaluation

We analyzed the effectiveness of VPNs and ad blockers by analyzing differences across the four experimental conditions (vanilla mode, VPN only, adblock only, and VPN plus adblock) based on data collected from 540 randomly selected websites. Our analysis focused on comparing the total number of third-party HTTP(S) requests, cookie storage events, JavaScript API calls, fingerprinting events, and cookie synchronizations, which ultimately relates to quantitative measures of online tracking activity.

4.1. 3rd-Party Requests per Site

(a)



(b)

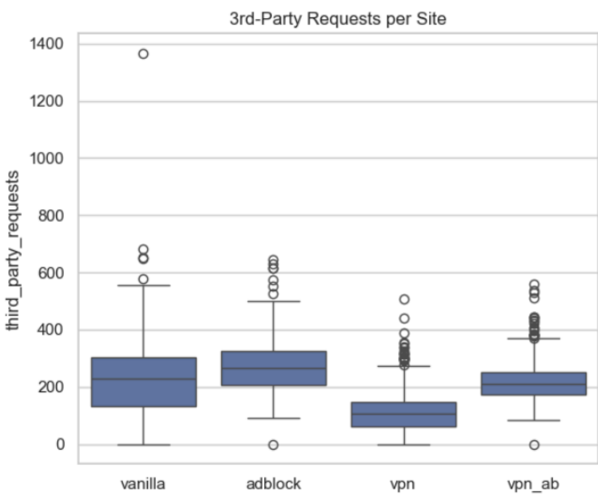


Figure 1: presents a comparison of the number of third-party requests per site across four browsing modes (vanilla, adblock, VPN, VPN + adblock) using (a) NordVPN (GUI) and (b) ProtonVPN (CLI), respectively.

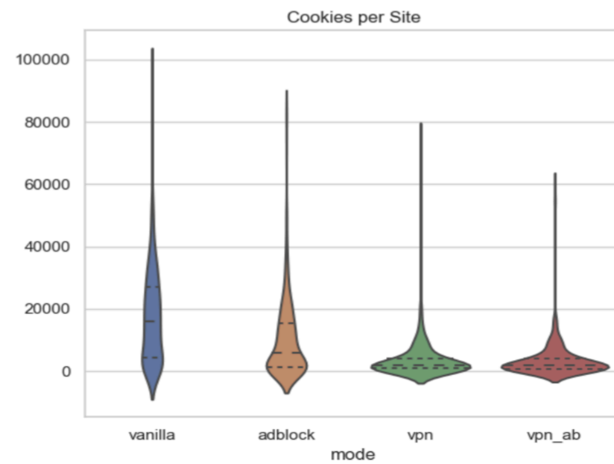
The number and distribution of third-party HTTP(S) requests per site under our four experimental conditions are shown in Figures 1(a) and 1(b) for NordVPN (GUI) and ProtonVPN (CLI), respectively. Overall, under the vanilla condition, NordVPN median site-level third-party requests are approximately 235, and ProtonVPN, 225. In the adblock condition, we see a 36% reduction with NordVPN (median ≈ 150 requests), and in the ProtonVPN experiment, there is a slight 11% increase in the amount of third-party requests (median ≈ 250 requests). This suggests some variability may have been introduced by the random routing of the nodes or timing of the measurement. In the VPN only condition, NordVPN provides a 15% reduction (median ≈ 200) and ProtonVPN via CLI provides significant approach a 56% reduction (median ≈ 100). Finally, the VPN + adblock produced median site-level third-party requests of 190 (NordVPN, 19% decrease) and 200 (ProtonVPN, 11% decrease) across sites.

These offsetting trends show that ad blocking in a GUI-based VPN client is the most significant influence on external requests, while a CLI-based VPN could itself substantially limit third-party connections, possibly as a function of routing, packet handling, or timing of request interception. Together, both tools achieved the largest overall impact for NordVPN, but only a modest further impact on ProtonVPN, emphasizing the client implementation was also important to the effectiveness of tracking reduction in the real world.

4.2. Cookies per Site

We investigated the long-term potential for client-side tracking by counting the total number of cookies that each origin set per site when a client visited that origin. Figures 2, (a) and (b) show the cookie count data plotted into the same four browsing modes and VPN clients.

(a)



(b)

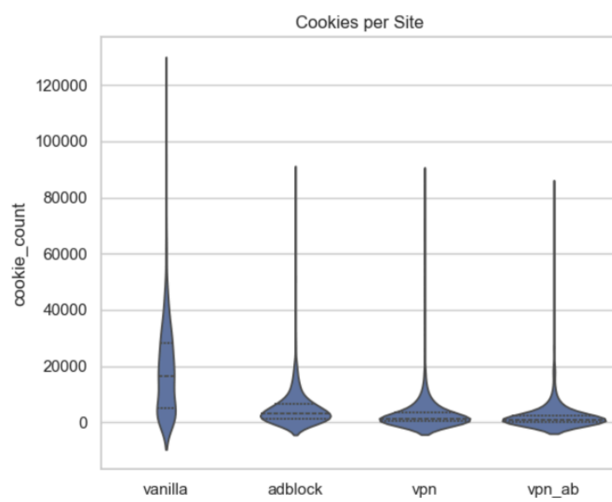
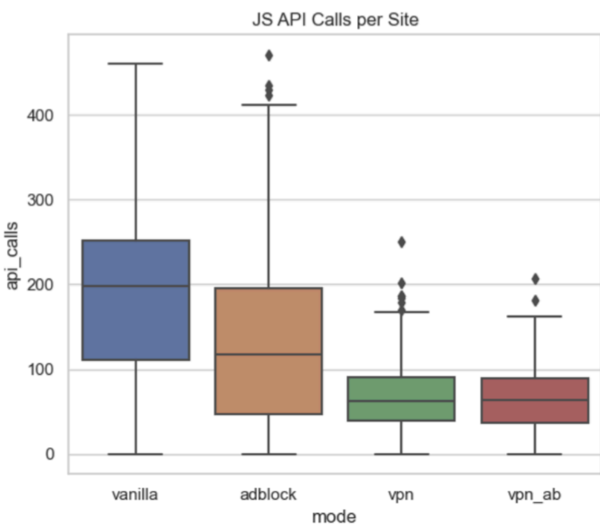


Figure 2: presents a comparison of the number of cookies set per site across four browsing modes (vanilla, adblock, VPN, VPN + adblock) using (a) NordVPN (GUI) and (b) ProtonVPN (CLI), respectively.

In both the GUI and CLI, vanilla mode has the largest median cookie counts and the most variability, which is based on a few sites issuing extraordinarily large quantities of cookies (in the NordVPN GUI test, one site exceeds 100,000 cookies). The introduction of adblock reduces the distribution considerably: in the GUI test, the median declines about 45 %–50 %, and in the CLI test about 75 %, showing that uBlock Origin is good at catching cookie-based trackers. In the VPN-only condition, the NordVPN GUI median reduces by almost 85 % and the ProtonVPN CLI median drops over 90 %, indicating that network-based routing and filtering are inhibiting cookie exchanges. Finally, VPN + adblock produces the lowest cookie counts overall–even though the additional reduction is small over adblock alone–indicating the highest level of protection occurs by using both.

4.3. JavaScript API Calls per Site

(a)



(b)

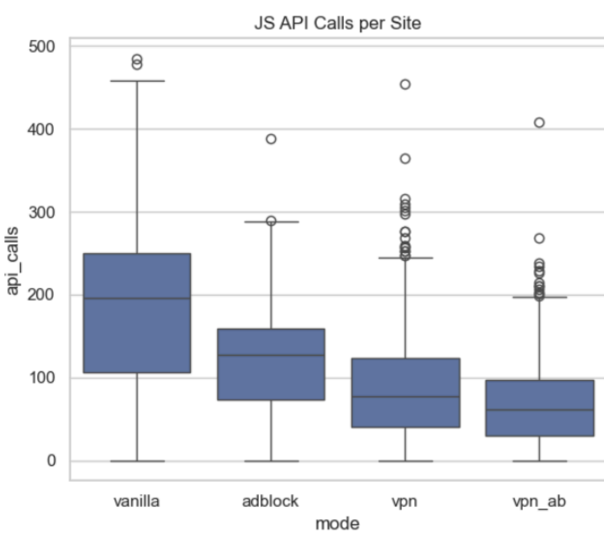
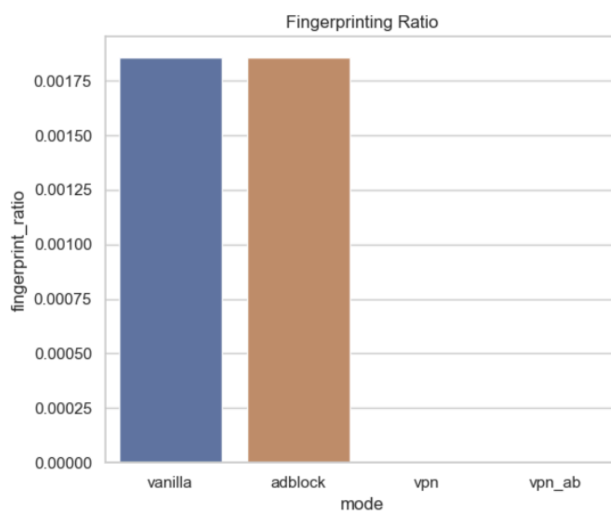


Figure 3: presents a comparison of the number of JavaScript API calls per site across four browsing modes (vanilla, adblock, VPN, VPN + adblock) using (a) NordVPN (GUI) and (b) ProtonVPN (CLI), respectively.

The distributions of API calls per site with NordVPN (GUI) and ProtonVPN (CLI) in the four browsing modes are illustrated in figures 4a and 4b below. Under vanilla mode, there are the highest median API calls (≈200 calls/site for both conditions) and large interquartile ranges, and lots of outliers—indicating that many individual sites are heavily invoking tracking-related APIs. Adding adblock reduced the median calls to approximately 40 % lower (NordVPN) and 35 % lower (ProtonVPN), illustrating uBlock Origin's ability to block or interrupt script-based tracking. The VPN-only condition resulted in further diminished calls: medians of ≈75 calls/site (NordVPN) and ≈70 calls/site (ProtonVPN)—showing that routing network traffic also can halt some tracking scripts. Finally, the combined VPN + adblock condition resulted in the lowest number of calls (≈60 calls/site GUI, ≈55 calls/site CLI) with the smallest interquartile ranges—but relatively little incremental benefit over adblock only.

4.4. Fingerprinting Ratio

(a)



(b)

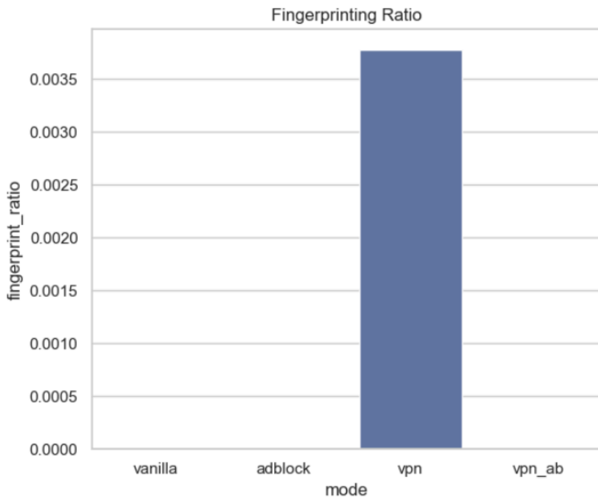


Figure 4: presents a comparison of the fingerprinting ratio across four browsing modes (vanilla, adblock, VPN, VPN + adblock) using (a) NordVPN (GUI) and (b) ProtonVPN (CLI), respectively.

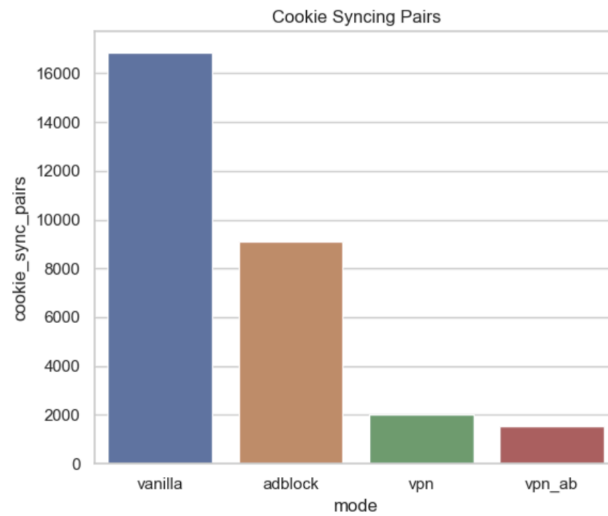
Evaluated the fingerprinting ratio across the four browsing modes to access how often websites attempted to uniquely identify users. As shown in Figures 4(a), and Figure 4(b) show that NordVPN (GUI) and ProtonVPN (CLI) setups exhibit different fingerprinting patterns.

The fingerprinting ratio measured in vanilla browsing for NordVPN (GUI) reached approximately 0.00175. The ad blocker implementation resulted in a minor decrease of fingerprinting ratio to 0.0017 while showing that this configuration did not substantially affect fingerprinting attempts. Network level protections seemed to block fingerprinting related requests because fingerprinting ratios remained negligible or were not recorded under VPN only and VPN with ad blocker condition.

ProtonVPN(CLI) demonstrated an entirely different fingerprinting behavior pattern. The fingerprinting ratio reached zero during both vanilla and ad block only testing period. The fingerprinting ratio of NordVPN's vanilla mode is nearly double. Exhibited unexpected results, which may be due to changes in traffic patterns or measurement randomness, rather than direct causation by VPN itself. The fingerprinting ratio dropped to zero when ProtonVPN users employed both VPN and ad blocker tools together in the VPN ad block mode. The research indicates that ad blockers failed to reduce fingerprinting under NordVPN while VPN usage without additional tools in ProtonVPN's CLI setup potentially increased fingerprinting exposure. Both VPN and ad blocker tools together provided the most effective protection throughout all testing scenarios.

4.5. Cookie Syncing Pairs

(a)



(b)

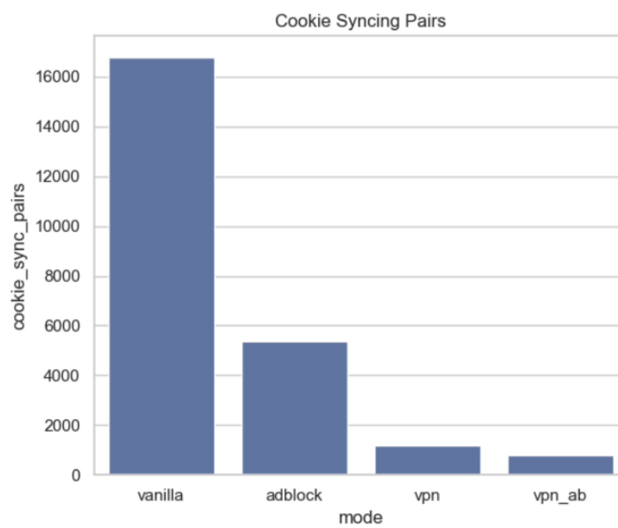


Figure 5: presents a comparison of the total number of cookie syncing pairs observed across four browsing modes (vanilla, adblock, VPN, VPN + adblock) using (a) NordVPN (GUI) and (b) ProtonVPN (CLI), respectively.

Under vanilla mode conditions, cookie synchronization is abundant, with approximately 17,000 sync pairs observed in both GUI and CLI experiments. In the adblock condition, the cookie sync was reduced by half (~9,000 pairs for NordVPN; ~5,300 pairs for ProtonVPN), indicating that uBlock Origin intercepts many of the sync requests. The VPN-only condition taxed the sync requests down further with a minimum of ~2,000 sync pairs (GUI) and ~1,200 sync pairs (CLI); indicating that network-level routing with the VPN blocks most of the sync requests which would initiate the exchange of cookies across domains. Finally, the VPN + adblock combination presents the least amount of syncs (~1,500 pairs GUI; ~800 pairs CLI) although the incremental gain from the VPN alone is modest. These results demonstrate that browser-level adblocking significantly reduces cookie syncing, and VPNs cause the most pourpoint reduction with a single tool, but the greatest privacy will result from using both tools in combination.

5. Ethics

Our study only examined publicly available websites while avoiding any contact with private user information. Research did not acquire or store or analyze any personal information throughout the experimental period. All crawling activities followed ethical standards rate limits to prevent website overload and the visits were designed to mimic typical human browning patterns. Our operation followed ethical standards for web measurement studies while maintaining complete absence of malicious activities and unauthorized access to vulnerabilities throughout data collection. All VPN usage occurred through valid subscription and the ad blocking configuration employed public software and filter lists.

6. Conclusion

The research aimed to determine VPN effectiveness and ad blocker capabilities in stopping online tracking activities when users browse the internet through Chrome. Our research analyzed 540 randomly chosen websites to evaluate the privacy effects of different browsing modes and VPN deployment types by examining third party requests, cookies, Javascript API calls, fingerprinting activities and cookie syncing.

The research demonstrates that VPNs succeed in concealing IP addresses and blocking some network level tracking methods yet fail to stop browser level tracking methods including fingerprinting and cookie setting. The essential role of uBlock Origin as an ad blocker became tracking protection. VPN protection together with ad blockers provided the most effective defense against all tracking activities. The analysis between NordVPN's GUI based and ProtonVPN's CLI based VPN showed significant differences between them. The CLI setup of ProtonVPN resulted in better third party requests and cookie reduction because it limited background system activities.

The ProtonVPN, VPN only condition showed unexpected growth in fingerprinting activity which indicates that VPN deployment methods together with traffic characteristics determine exposure to particular tracking methods. The research demonstrates that complete privacy demands multiple defensive layers because VPNs fail to stop contemporary tracking methods. Users need to implement VPNs for network level protection alongside ad blockers for browser level defense to achieve complete privacy protection. Future research should extend this study by examining different browsers, mobile platforms, and various VPN providers to determine privacy effects in multiple environments.

7. Reference

[1] Allied Telesis. (n.d.). *Getting started with the device GUI on VPN routers*. Allied Telesis. https://www.alliedtelesis.com/sites/default/files/documents/getting-started-guides/getting_started_with_the_device_gui_on_vpn_routers.pdf

[2] Gunes Acar, Marc Juarez, Nick Nikiforakis, et al. "OpenWPM: An Automated Platform for Web Privacy Measurement." Proceedings of the 2016 ACM SIGSAC ConferenceonComputerandCommunicationsSecurity(CCS' 16). https://dl.acm.org/doi/10.1145/2976749.2978313

[3] Electronic Frontier Foundation (EFF). "Panopticlick Project." https://panopticlick.eff.org/

[4] DuckDuckGo Tracker Radar Dataset. GitHub Repository. https://github.com/duckduckgo/tracker-radar

[5] Ikram, M., Vallina-Rodriguez, N., Seneviratne, S., Kaafar, M. A., & Paxson, V. (2016). Analyzing the Privacy and Security Risks of Android VPN Permission-enabled Apps. *Proceedings of the 2016 Internet Measurement Conference (IMC '16)*, 349–364.

[6] N. Vallina-Rodriguez, J. Shah, A. Finamore, K. Papagiannaki, Y. Grunenberger, and J. Crowcroft, "A Glance through the VPN Looking Glass: IPv6 Leakage and DNS Hijacking," Proceedings on Privacy Enhancing Technologies (PoPETs), 2017(2), pp. 77–91. https://petsymposium.org/2017/papers/issue2/paper47-2017-2-source.pdf

[7] A. Valls, A. Reyes-Menendez, and F. Martinez-Navarro, "Ad-Blocking and Anti-Tracking: A Systematic Literature Review," arXiv preprint arXiv:2410.23504, 2024. https://arxiv.org/pdf/2410.23504